# Distensibility Identification Formulation

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## 1 Steady Poiseuille Flow Network

### 1.1 Poiseuille Flow

The developing vasculature is modeled as a graph consisting of nodes and edges, where nodes represent vessel bifurcations or junctions and edges represent individual vessel segments. Each edge $e$, connecting nodes $i$ and $j$, is approximated as a cylindrical vessel with radius $R_e$ and length $L_e$. For all analyses, assume the vessel geometry, cardiac heart rate, fluid density, and fluid dynamic viscosity are known. Let $P_i$ and $P_j$ denote the pressures at nodes $i$ and $j$, respectively. The fluid is assumed to have constant density $\rho$ and dynamic viscosity $\mu$. Under the assumptions of steady, incompressible, Newtonian flow, the volumetric flow rate through vessel segment $e$ is described by Poiseuille's law,

$$Q_e = Q_{ij} = \frac{\pi R_e^4}{8\mu L_e}(P_i - P_j) = g_e(P_i - P_j) = g_{ij}(P_i - P_j) \tag{1}$$

where

$$g_e = g_{ij} = \frac{\pi R_e^4}{8\mu L_e} \tag{2}$$

is the hydraulic conductance.

### 1.2 Flow Network

Neighboring vessels are coupled through the pressures at shared nodes. The nodal pressures are determined by enforcing conservation of mass throughout the network. For node $i$, conservation of mass requires that the net flow into or out of the node balance any externally imposed source or sink term,

$$\sum_j Q_{ij} = H_i, \tag{3}$$

where the sum is taken over all nodes $j$ connected to node $i$ and $H_i$ denotes the net external flow applied at node $i$. A positive $H_i$ represents fluid injected into the network (a source), a negative $H_i$ represents fluid removed from the network (a sink), and $H_i = 0$ corresponds to an interior node with no external inflow or outflow. Substituting equation 1 into equation 3 yields

$$\sum_j g_{ij}(P_i - P_j) = \left(\sum_j g_{ij}\right) P_i - \sum_j g_{ij} P_j = H_i. \tag{4}$$

In matrix form, for a network consisting of $N$ nodes, equation 4 can be written as

$$\mathbf{LP} = \mathbf{H} \tag{5}$$

where

$$\mathbf{P} = (P_1, P_2, ..., P_N)^T$$

contains the nodal pressures and

$$\mathbf{H} = (H_1, H_2, ..., H_N)^T$$

contains net external flows. The matrix $\mathbf{L}$ is the *weighted graph Laplacian* which encodes both the vascular topology and the hydraulic conductances of the vessel segments, thereby coupling pressures throughout the network. Each vessel contributes a local conductance matrix to the global system. Specifically, for a vessel segment $e$ connecting nodes $i$ and $j$ with conductance $g_e$, these contributions to $\mathbf{L}$ are stamped as

$$\begin{aligned} L_{ii} &\leftarrow L_{ii} + g_e, \qquad & L_{jj} &\leftarrow L_{jj} + g_e, \\ L_{ij} &\leftarrow L_{ij} - g_e, \qquad & L_{ji} &\leftarrow L_{ji} - g_e. \end{aligned}$$

The global Laplacian matrix is obtained by summing these contributions over all vessel segments in the network.

### 1.3 Limitations of the Steady Model

The steady Poiseuille flow network model treats each vessel as a rigid hydraulic resistor. While this approximation captures the mean flow distribution throughout the network, it neglects deformation of the vessel wall in response to pressure. Pressure and flow vary throughout the cardiac cycle. As pressure increases, compliant vessels expand and temporarily store fluid; as pressure decreases, this stored fluid is released back into the circulation. Consequently, flow is not transmitted instantaneously through the network. Instead, pulsatile pressure and flow waves experience attenuation and phase delays as they propagate through compliant vessels.

These effects cannot be represented by a purely steady model. Capturing the dynamic relationship between pressure and flow requires a time-dependent description that accounts for vessel compliance. We therefore extend the formulation to a distributed model of flow in deformable vessels. A key quantity governing this behavior is the vessel distensibility.

## 2 Distensibility

### 2.1 Power Law Formulation

Consider the cross section of a vessel segment shown in Figure 1. Under the thin-wall approximation, Laplace's law gives

$$\sigma_\theta \sim \frac{P_{tm} R_e}{h},$$

where $\sigma_\theta$ is the circumferential wall stress, $P_{tm}$ is the transmural pressure, and $h$ is the wall thickness. Assuming the vessel wall is linearly elastic, Hooke's law yields

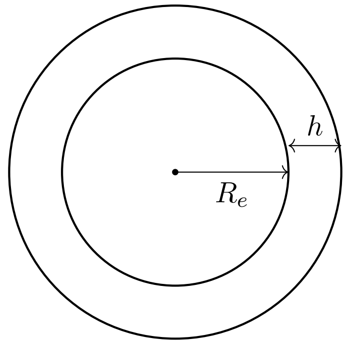


Figure 1: Cross-sectional geometry of a vessel segment with lumen radius $R_e$ and wall thickness $h$.

$$\epsilon_\theta \sim \frac{\sigma_\theta}{E} \sim \frac{P_{tm} R_e}{Eh},$$

where $\epsilon_\theta$ is the circumferential strain and $E$ is the Young's modulus. For small deformations, the circumferential strain is

$$\epsilon_\theta = \frac{\Delta R_e}{R_e}.$$

Since the cross-sectional area is $A_e = \pi R_e^2$,

$$\frac{\Delta A_e}{A_e} = \frac{\pi (R_e + \Delta R_e)^2 - \pi R_e^2}{\pi R_e^2} \approx 2\frac{\Delta R_e}{R_e},$$

where $\Delta R_e$ is small and terms of order $(\Delta R_e)^2$ have been neglected. Therefore,

$$\epsilon_\theta \approx \frac{1}{2}\frac{\Delta A_e}{A_e}.$$

where $\epsilon_\theta$ is the wall strain and $E$ is the Young's modulus. Assuming the vessel length $L_e$ remains approximately constant, the distensibility is

$$D_e = \frac{\partial_P V_e}{V_e} = \frac{\partial_P A_e}{A_e}$$

Using the strain scaling derived above[1] and assuming $\Delta P_{tm} \approx \Delta P$,

$$D_e \sim \frac{\Delta A_e}{A_e \Delta P} \sim \frac{R_e}{Eh}$$

Assuming $h \sim R_e$, then $D_e \sim E^{-1}$ and is radius-independent. More generally, to allow for radius-dependent stiffness and/or deviations from geometric similarity, we consider a power law,

$$D_e(R_e) = D_0 \left(\frac{R_e}{R_0}\right)^\alpha \tag{6}$$

where $R_0$ is a characteristic reference radius, $D_0$ is the distensibility at $R_e = R_0$, and $\alpha$ controls the radius dependence. Note that $D_e$ is now edge-dependent. Setting $\alpha = 0$ recovers constant distensibility.

[1]The preceding stress–strain relations describe the vessel response to a transmural pressure load. Distensibility is an incremental quantity, defined in terms of the area change produced by a pressure perturbation. Consequently, $P$ is replaced by a small pressure increment $\Delta P$ when relating area changes to distensibility.

### 2.2 Selecting the characteristic reference radius $R_0$

The choice of $R_0$ is arbitrary and serves only to nondimensionalize the radius. Different choices of $R_0$ lead to a corresponding rescaling of $D_0$ while leaving the predicted distensibility unchanged. Choosing $R_0$ near the center of the observed radius distribution may aid in numerical stability when fitting $D_0$ and/or $\alpha$.

### 2.3 Selecting or determining $\alpha$

Given $R_0$, the remaining parameters to determine are $D_0$ and $\alpha$. Selecting $\alpha$ allows exploration of radius-independent parameters. Solving for $\alpha$ can test hypotheses for $D_e$'s dependence on $R_e$.

## 3 Distributed Pressure Compliance

### 3.1 Conservation Equations

For pulsatile flow, the pressure and flow vary in space and time,

$$P = P(z,t), \qquad Q = Q(z,t).$$

Further, the compliance per unit length is

$$c_e = \frac{\partial A_e}{\partial P} = A_e D_e = \pi D_0 \left( \frac{R_e^{\alpha+2}}{R_0^{\alpha}} \right) \tag{7}$$

Strictly speaking, the compliance is time-dependent. We assume that compliance is approximately constant. Of note, setting $\alpha = -2$ makes $c_e$ independent of vessel radius.

#### 3.1.1 Mass Conservation

Any imbalance of flow into a vessel segment must change the stored fluid volume. Assuming the length of a vessel, $L_e$, remains constant,

$$\partial_t A_e + \partial_z Q_e = 0 \tag{8}$$

With constant compliance, to first-order, we model the changes in area as

$$A_e(P) \approx A_0 + (P - P_0)c_e$$

where $A_0$ and $P_0$ are reference area and pressure, respectively. Thus,

$$\partial_t A_e = c_e \partial_t P$$

Importantly, the term $c_e \partial_t P$ is absent from Poiseuille flow and represents local fluid storage in the vessel wall. The mass conservation equation is now

$$c_e \partial_t P + \partial_z Q_e = 0 \tag{9}$$

#### 3.1.2 Momentum Conservation

Consider the axial component of the Navier-Stokes equation and assume 1-dimensional flow parallel to the axis. The pressure, viscous, and inertial terms are of interest:

$$\partial_z P + r_e Q_e + \ell_e \partial_t Q_e = 0 \tag{10}$$

where

$$r_e = \frac{1}{g_e L_e}, \qquad \ell_e = \frac{\rho}{\pi R_e^2}.$$

$r_e$ is the hydraulic resistance per unit length and $\ell_e$ is the intertance per unit length. Importantly, in the steady limit (i.e., $\partial_t Q = 0$), the momentum balance for Poiseuille flow is recovered.

We nondimensionalize equation 10 using the following relations:

$$\tilde{z} = \frac{z}{L_e}, \qquad \tilde{t} = \omega_0 t = 2\pi f_0 t, \qquad \tilde{Q}_e = \frac{Q_e}{Q_0}, \qquad \tilde{P} = \frac{P}{P_0} = \frac{P}{r_e L_e Q_0}$$

Here, $f_0$ is the fundamental heartbeat frequency, $Q_0$ is a characteristic flow scale, and $P_0$ is the corresponding viscous pressure scale[2]. The nondimensional momentum balance is now,

$$\partial_{\tilde{z}} \tilde{P} + \tilde{Q}_e + \frac{\ell_e \omega_0}{r_e} \partial_{\tilde{t}} \tilde{Q}_e = 0 \tag{11}$$

The Womersley Number is

$$Wo_e^2 = \frac{8 \ell_e \omega_0}{r_e} = \frac{\rho \omega_0 R_e^2}{\mu} \tag{12}$$

In embryonic vessels, inertial effects are negligible and $Wo_e \ll 1$. Dropping the inertial term in equation 11 and re-introducing dimensions yields

$$\partial_z P + r_e Q_e = 0 \tag{13}$$

#### 3.1.3 Diffusion Equation

Equations 9 and 13 for a coupled system of first-order partial differential equations. Assuming $r_e$ is approximately constant, solving for $Q$ in equation 13 and substituting in equation 9 yields

$$\frac{\partial_z^2 P}{r_e c_e} = \partial_t P \tag{14}$$

Equation 14 has the form of a diffusion equation, with hydraulic diffusivity

$$D_{\mathrm{hyd,e}} = \frac{1}{r_e c_e} = \frac{R_0^\alpha}{8\mu D_0} R_e^{2-\alpha}.$$

Thus, in the low-inertia limit, pulsatile pressure disturbances do not propagate as undamped waves. Instead, they spread diffusively through the compliant vessel, with viscous resistance dissipating

[2] The characteristic flow scale $Q_0$ may be chosen as a representative mean (DC) flow magnitude. The corresponding pressure scale $P_0 = r_e L_e Q_0$ is the steady Poiseuille pressure drop required to drive the reference flow through a vessel segment. However, in this formulation, we are motivating why the inertial term in equation 10 may be dropped and the exact choices of $Q_0$ and $P_0$ are not critical for the present analysis.

the signal and wall compliance providing local fluid storage. Over a characteristic timescale $\omega_0^{-1} = (2\pi f_0)^{-1}$, the distance over which a pressure disturbance can diffuse is

$$\delta_{hyd} = \sqrt{\frac{R_0^\alpha R_e^{2-\alpha}}{16\pi\mu D_0 f_0}}$$

We refer to $\delta_{hyd}$ as the *hydraulic diffusion length.* It represents the characteristic distance over which pulsatile pressure variations can equilibrate during a single heartbeat. Setting $\alpha = 2$ causes both the hydraulic diffusivity and the hydraulic diffusion length to become independent of vessel radius.

To compare the vessel length to the hydraulic diffusion length, we define the dimensionless edge compliance number

$$\mathcal{C}_e = \left(\frac{L_e}{\delta_{hyd}}\right)^2 \tag{15}$$

When $\mathcal{C}_e \ll 1$, the hydraulic diffusion length greatly exceeds the vessel length, and pressure has sufficient time to equilibrate across the vessel within a single cardiac cycle. In this regime, pulsatile pressure variations are transmitted with little attenuation. This regime is appropriate for the developing vasculature. Conversely, when $\mathcal{C}_e \gg 1$, the vessel length exceeds the hydraulic diffusion length, and pressure disturbances decay substantially before traversing the vessel. As a result, pulsatile signals are strongly attenuated. Notably, higher frequencies attenuate more rapidly and accumulate larger phase lags.

## 3.2 Network Assembly for Pulsatile Harmonics

### 3.2.1 Solving for Pressure

For steady flow, each edge contributes a real conductance $g$ to the weighted graph Laplacian. For pulsatile flow, for each harmonic $n$, each edge contributes complex, frequency-dependent admittances. We rewrite pressure as a harmonic phasor,

$$P(z,t) = \hat{P}^{(n)}(z)\exp(in\omega_0 t) \tag{16}$$

Substituting equation 16 into equation 14 gives, for each harmonic $n$,

$$\partial_z^2 \hat{P}^{(n)} = in\omega_0 r_e c_e \hat{P}^{(n)} = \left(k_e^{(n)}\right)^2 \hat{P}^{(n)}. \tag{17}$$

We define the complex hydraulic wavenumber

$$k_e^{(n)} = \sqrt{in\omega_0 r_e c_e} = \sqrt{\frac{in\omega_0}{D_{\text{hyd},e}}} = \frac{\sqrt{i\mathcal{C}_e^{(n)}}}{L_e} \propto R_e^{(\alpha-2)/2},$$

where the square-root branch is chosen so that $\Re k_e^{(n)} > 0$. $k_e^{(n)}$ measures the rate of harmonic attenuation and phase accumulation along edge $e$.
Let the endpoint pressure phasors be

$$\hat{P}^{(n)}(0) = \hat{P}_i^{(n)}, \qquad \hat{P}^{(n)}(L_e) = \hat{P}_j^{(n)}.$$

The solution of equation 17 satisfying these endpoint conditions is

$$\hat{P}^{(n)}(z) = \hat{P}_i^{(n)} \frac{\sinh\left(k_e^{(n)}(L_e - z)\right)}{\sinh\left(k_e^{(n)} L_e\right)} + \hat{P}_j^{(n)} \frac{\sinh\left(k_e^{(n)} z\right)}{\sinh\left(k_e^{(n)} L_e\right)}. \tag{18}$$

### 3.2.2 Solving for Flow

Substituting equation 18 into equation 13 gives

$$\begin{aligned} \hat{Q}_e^{(n)} = \hat{Q}_{i\to j}^{(n)} &= \frac{k_e^{(n)}}{r_e}\left[\hat{P}_i^{(n)} \coth\left(k_e^{(n)} L_e\right) - \hat{P}_j^{(n)} \operatorname{csch}\left(k_e^{(n)} L_e\right)\right] \\ &= Y_{s,e}^{(n)} \hat{P}_i^{(n)} - Y_{t,e}^{(n)} \hat{P}_j^{(n)} = Y_{s,ij}^{(n)} \hat{P}_i^{(n)} - Y_{t,ij}^{(n)} \hat{P}_j^{(n)}. \end{aligned} \tag{19}$$

Here,

$$Y_{s,e}^{(n)} = \frac{k_e^{(n)}}{r_e} \coth\left(k_e^{(n)} L_e\right), \qquad Y_{t,e}^{(n)} = \frac{k_e^{(n)}}{r_e} \operatorname{csch}\left(k_e^{(n)} L_e\right)$$

are the self- and trans-admittances, respectively.

Consider the case where $|k_e^{(n)} L_e| \ll 1$ or equivalently, $\mathcal{C}_e^{(n)} \ll 1$. In this limit, the edge is short relative to the harmonic hydraulic diffusion length, so pressure nearly equilibrates across the edge during one oscillation. The self- and trans-admittances become

$$Y_{s,e}^{(n)} \approx g_e\left(1 + \frac{(k_e^{(n)} L_e)^2}{3}\right) \qquad Y_{t,e}^{(n)} \approx g_e\left(1 - \frac{(k_e^{(n)} L_e)^2}{6}\right)$$

Consequently, each vessel is essentially a resistor carrying a small pressure-storage phase shift. As $|k_e^{(n)} L_e| \to 0$ (equivalently, $\mathcal{C}_e^{(n)} \to 0$), both admittances return to $g_e$ and the steady network is recovered.

## 3.3 Flow Network with Harmonics

At each node, the sum of outgoing harmonic flows must balance any imposed harmonic injection,

$$\sum_j \hat{Q}_{i\to j}^{(n)} = \hat{H}_i^{(n)} \tag{20}$$

Substituting equation 19 into equation 20 yields

$$\sum_j \left(Y_{s,ij}^{(n)} \hat{P}_i^{(n)} - Y_{t,ij}^{(n)} \hat{P}_j^{(n)}\right) = \hat{H}_i^{(n)} \tag{21}$$

In matrix form, equation 21 can be written as

$$\mathbf{Y}^{(n)} \hat{\mathbf{P}}^{(n)} = \hat{\mathbf{H}}^{(n)} \tag{22}$$

where for the $n^{th}$ harmonic,

$$\hat{\mathbf{P}}^{(n)} = (\hat{P}_1^{(n)}, \hat{P}_2^{(n)}, ..., \hat{P}_N^{(n)})^T$$

contains the nodal pressures and

$$\hat{\mathbf{H}}^{(n)} = (\hat{H}_1^{(n)}, \hat{H}_2^{(n)}, ..., \hat{H}_N^{(n)})^T$$

contains net external flows. The matrix $\mathbf{Y}^{(n)}$ is the complex harmonic nodal matrix, which encodes both the vascular topology and the frequency-dependent hydraulic admittances of the vessel segments, thereby coupling pressure amplitudes and phases throughout the network. Each vessel contributes a local admittance matrix to the global system. Specifically, for a vessel segment $e$ connecting nodes $i$ and $j$ with self-admittance $Y_{s,e}^{(n)}$ and trans-admittance $Y_{t,e}^{(n)}$, these contributions are stamped as

$$\begin{aligned} Y_{ii}^{(n)} &\leftarrow Y_{ii}^{(n)} + Y_{s,e}^{(n)}, \\ Y_{jj}^{(n)} &\leftarrow Y_{jj}^{(n)} + Y_{s,e}^{(n)}, \\ Y_{ij}^{(n)} &\leftarrow Y_{ij}^{(n)} - Y_{t,e}^{(n)}, \\ Y_{ji}^{(n)} &\leftarrow Y_{ji}^{(n)} - Y_{t,e}^{(n)}. \end{aligned}$$

The global harmonic nodal matrix is obtained by summing these contributions over all vessel segments in the network.

In the limit $|k_e^{(n)} L_e| \to 0$, the self- and trans-admittances both approach the Poiseuille conductance $g_e$, and $\mathbf{Y}^{(n)}$ reduces to the weighted graph Laplacian $\mathbf{L}$ used in the steady-flow formulation.

# 4 Simulation and Conversion to Velocity Formulation

The harmonic network equations define a forward simulation that maps specified boundary pressure phasors to predicted edge-flow and, as will be shown in section 4.2, edge-velocity phasors. Importantly, the distensibility parameters must be specified before the harmonic network operator is defined. For a prescribed reference radius $R_0$ and fixed $(D_0, \alpha)$[3], each edge has a specified distensibility $D_e$, compliance per unit length $c_e$, hydraulic wavenumber $k_e^{(n)}$, and harmonic admittances $Y_{s,e}^{(n)}$ and $Y_{t,e}^{(n)}$. These edge admittances define the global harmonic nodal matrix $\mathbf{Y}^{(n)}(D_0, \alpha)$. The boundary pressure phasors then specify how the network is driven.

Therefore, for each harmonic $n$, the forward simulation is defined as follows: given the network geometry, fluid properties, driving frequency, distensibility parameters, and boundary pressure phasors $\hat{\mathbf{P}}_b^{(n)}$, solve for the interior pressure phasors and then compute the corresponding edge-flow and edge-velocity phasors.

The harmonic nodal system is

$$\mathbf{Y}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}^{(n)} = \hat{\mathbf{H}}^{(n)}. \tag{23}$$

Partition the pressure phasors into interior nodes (int) and boundary nodes ($b$),

$$\hat{\mathbf{P}}^{(n)} = \begin{pmatrix} \hat{\mathbf{P}}_{\text{int}}^{(n)} \\ \hat{\mathbf{P}}_b^{(n)} \end{pmatrix},$$

and partition the global harmonic nodal matrix accordingly,

$$\mathbf{Y}^{(n)}(D_0, \alpha) = \begin{pmatrix} \mathbf{Y}_{\text{int,int}}^{(n)}(D_0, \alpha) & \mathbf{Y}_{\text{int},b}^{(n)}(D_0, \alpha) \\ \mathbf{Y}_{b,\text{int}}^{(n)}(D_0, \alpha) & \mathbf{Y}_{b,b}^{(n)}(D_0, \alpha) \end{pmatrix}.$$

[3]Equivalently, for fixed $\alpha$, one may specify the combined coefficient $D_0/R_0^{\alpha}$.

The interior equations are

$$\mathbf{Y}^{(n)}_{\text{int,int}}(D_0,\alpha)\hat{\mathbf{P}}^{(n)}_{\text{int}} + \mathbf{Y}^{(n)}_{\text{int},b}(D_0,\alpha)\hat{\mathbf{P}}^{(n)}_{b} = \hat{\mathbf{H}}^{(n)}_{\text{int}}. \tag{24}$$

Solving for the unknown interior pressure phasors gives

$$\hat{\mathbf{P}}^{(n)}_{\text{int}} = \left(\mathbf{Y}^{(n)}_{\text{int,int}}(D_0,\alpha)\right)^{-1}\left[\hat{\mathbf{H}}^{(n)}_{\text{int}} - \mathbf{Y}^{(n)}_{\text{int},b}(D_0,\alpha)\hat{\mathbf{P}}^{(n)}_{b}\right]. \tag{25}$$

In the common case where there are no imposed interior harmonic injections,

$$\hat{\mathbf{H}}^{(n)}_{\text{int}} = \mathbf{0},$$

equation 25 reduces to

$$\hat{\mathbf{P}}^{(n)}_{\text{int}} = -\left(\mathbf{Y}^{(n)}_{\text{int,int}}(D_0,\alpha)\right)^{-1}\mathbf{Y}^{(n)}_{\text{int},b}(D_0,\alpha)\hat{\mathbf{P}}^{(n)}_{b}. \tag{26}$$

This shows that, for fixed distensibility parameters $(D_0,\alpha)$, the interior pressure phasors depend linearly on the specified boundary pressure phasors.

## 4.1 Solving for Flow Harmonics

Once the pressure phasors are known, the harmonic volumetric flow through edge $e = (i,j)$ is obtained from the edge admittance relation,

$$\hat{Q}^{(n)}_{e} = Y^{(n)}_{s,e}(D_0,\alpha)\hat{P}^{(n)}_{i} - Y^{(n)}_{t,e}(D_0,\alpha)\hat{P}^{(n)}_{j}. \tag{27}$$

Since the pressure solution is linear in $\hat{\mathbf{P}}^{(n)}_{b}$, the edge-flow phasors are also linear in $\hat{\mathbf{P}}^{(n)}_{b}$. Therefore, for fixed $(D_0,\alpha)$, the network defines a harmonic flow-transfer matrix,

$$\hat{\mathbf{Q}}^{(n)} = \mathbf{T}^{(n)}_{Q}(D_0,\alpha)\hat{\mathbf{P}}^{(n)}_{b}. \tag{28}$$

The matrix $\mathbf{T}^{(n)}_{Q}(D_0,\alpha)$ is constructed by solving the harmonic network problem for a set of basis boundary pressure phasors and reading out the corresponding edge-flow phasors using equation 27. Thus, $(D_0,\alpha)$ determine the harmonic network operator, while $\hat{\mathbf{P}}^{(n)}_{b}$ determines the boundary driving.

## 4.2 Solving for Velocity Harmonics

The measured quantity is often the velocity rather than volumetric flow. Thus, it is worthwhile to reformulate solving for the flow harmonics to solving for the velocity harmonics[4]

### 4.2.1 The Zeroth Harmonic: DC Flow

For the zeroth harmonic, the flow is steady or mean flow. The corresponding mean velocity is

$$v^{(0)}_{e} = \frac{Q^{(0)}_{e}}{\pi R_e^2} = \frac{g_e}{\pi R_e^2}(P^{(0)}_{i} - P^{(0)}_{j}) = \tilde{g}_e(P^{(0)}_{i} - P^{(0)}_{j}) \tag{29}$$

where

$$\tilde{g}_e = \frac{g_e}{\pi R_e^2} = \frac{R_e^2}{8\mu L_e} \tag{30}$$

is the conductivity scaled by area.

[4] We assume that the average edge radius $\overline{R}_e$ is approximately the same as $R_e$ for all edges. This is supported by the previous assumption in section 2.1 where $\Delta R_e$ is small.

### 4.2.2 Higher Harmonics: Pulsatile Flow

For higher harmonics ($n \geq 1$),

$$\hat{v}_e^{(n)} = \frac{\hat{Q}_e^{(n)}}{\pi R_e^2}. \tag{31}$$

Substituting equation 27 gives

$$\hat{v}_e^{(n)} = \frac{Y_{s,e}^{(n)}(D_0, \alpha)}{\pi R_e^2} \hat{P}_i^{(n)} - \frac{Y_{t,e}^{(n)}(D_0, \alpha)}{\pi R_e^2} \hat{P}_j^{(n)}.$$

Define the velocity admittances

$$\tilde{Y}_{s,e}^{(n)} = \frac{Y_{s,e}^{(n)}}{\pi R_e^2}, \qquad \tilde{Y}_{t,e}^{(n)} = \frac{Y_{t,e}^{(n)}}{\pi R_e^2}.$$

Then

$$\hat{v}_e^{(n)} = \tilde{Y}_{s,e}^{(n)}(D_0, \alpha) \hat{P}_i^{(n)} - \tilde{Y}_{t,e}^{(n)}(D_0, \alpha) \hat{P}_j^{(n)}. \tag{32}$$

Equivalently, if

$$\mathbf{A}_E = \operatorname{diag}\left(\pi R_1^2, \pi R_2^2, \ldots, \pi R_{N_e}^2\right)$$

is the diagonal matrix of edge cross-sectional areas for $N_e$ edges, then

$$\hat{\mathbf{v}}^{(n)} = \mathbf{A}_E^{-1} \hat{\mathbf{Q}}^{(n)}.$$

Combining this expression with equation 28, including the zeroth harmonic, gives the velocity-transfer form

$$\hat{\mathbf{v}}^{(n)} = \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha) \hat{\mathbf{P}}_b^{(n)}, \tag{33}$$

where

$$\widetilde{\mathbf{T}}^{(n)}(D_0, \alpha) = \mathbf{A}_E^{-1} \mathbf{T}_Q^{(n)}(D_0, \alpha). \tag{34}$$

Thus, the same pressure solve is used whether the simulation output is flow or velocity. The difference is only in the output map applied after solving for the pressure field. Volumetric conservation is enforced through the harmonic nodal matrix $\mathbf{Y}^{(n)}$, while velocity harmonics are obtained by dividing each edge-flow harmonic by the corresponding cross-sectional area.

## 4.3 Adding Measurement Noise

To generate synthetic observations for method testing, noise may be added to the simulated edge-flow or edge-velocity harmonics. Since the model output is complex-valued for each harmonic, noise is added independently to the real and imaginary components. For velocity measurements, we write

$$\hat{\mathbf{v}}_{\text{obs}}^{(n)} = \hat{\mathbf{v}}^{(n)} + \boldsymbol{\eta}_v^{(n)}, \tag{35}$$

where $\hat{\mathbf{v}}^{(n)}$ is the noiseless simulated velocity phasor and $\boldsymbol{\eta}_v^{(n)}$ is complex measurement noise. A simple additive Gaussian noise model is

$$\boldsymbol{\eta}_v^{(n)} = \boldsymbol{\eta}_{v,\text{Re}}^{(n)} + i\boldsymbol{\eta}_{v,\text{Im}}^{(n)}, \tag{36}$$

with

$$\boldsymbol{\eta}_{v,\mathrm{Re}}^{(n)} \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_{v,n}^2}{2}\mathbf{I}\right), \qquad \boldsymbol{\eta}_{v,\mathrm{Im}}^{(n)} \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_{v,n}^2}{2}\mathbf{I}\right).$$

Equivalently,

$$\boldsymbol{\eta}_v^{(n)} \sim \mathcal{CN}\left(\mathbf{0}, \sigma_{v,n}^2\mathbf{I}\right),$$

where $\mathcal{CN}$ denotes a circularly symmetric complex Gaussian distribution. The factor of 1/2 in the real and imaginary variances ensures that

$$\mathbb{E}\left[\left|\eta_{v,e}^{(n)}\right|^2\right] = \sigma_{v,n}^2 \tag{37}$$

for each edge $e$.

If the measurement uncertainty differs across edges, the noise covariance may instead be specified as

$$\boldsymbol{\eta}_v^{(n)} \sim \mathcal{CN}\left(\mathbf{0}, \boldsymbol{\Sigma}_v^{(n)}\right), \tag{38}$$

where $\boldsymbol{\Sigma}_v^{(n)}$ is typically chosen to be diagonal. For example, edge-dependent noise levels can be written as

$$\boldsymbol{\Sigma}_v^{(n)} = \mathrm{diag}\left(\sigma_{v,1,n}^2, \sigma_{v,2,n}^2, \ldots, \sigma_{v,N_e,n}^2\right). \tag{39}$$

The same construction may be applied to flow harmonics,

$$\hat{\mathbf{Q}}_{\mathrm{obs}}^{(n)} = \hat{\mathbf{Q}}^{(n)} + \boldsymbol{\eta}_Q^{(n)}. \tag{40}$$

In the simulations below, the noise model specifies how noiseless forward-model outputs are converted into synthetic observations. The underlying simulated quantities $\hat{\mathbf{P}}^{(n)}$, $\hat{\mathbf{Q}}^{(n)}$, and $\hat{\mathbf{v}}^{(n)}$ remain determined entirely by the prescribed geometry, fluid properties, distensibility parameters, and boundary pressure phasors.

## 4.4 Simulation Outputs

For each specified harmonic $n$, the forward simulation produces the nodal pressure phasors,

$$\hat{\mathbf{P}}^{(n)} = \begin{pmatrix} \hat{\mathbf{P}}_{\mathrm{int}}^{(n)} \\ \hat{\mathbf{P}}_b^{(n)} \end{pmatrix},$$

the edge-flow phasors,

$$\hat{\mathbf{Q}}^{(n)} = \mathbf{T}_Q^{(n)}(D_0, \alpha)\hat{\mathbf{P}}_b^{(n)},$$

and the edge-velocity phasors,

$$\hat{\mathbf{v}}^{(n)} = \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}_b^{(n)}.$$

These simulated harmonics may be used to reconstruct time-domain pressure, flow, or velocity signals by summing over the prescribed harmonics.

# 5 Inference

For inference, we only know the measured edge-velocity harmonics, the network geometry, and fluid density and dynamic viscosity. In particular, for each measured edge $e$ and harmonic $n$, we have

$$\hat{v}^{(n)}_{\text{obs},e}.$$

The distensibility parameters $(D_0, \alpha)$ are unknown, and the boundary pressure phasors $\hat{\mathbf{P}}^{(n)}_b$ are also unknown. They must be estimated from the measured velocity harmonics.

For fixed trial values of $(D_0, \alpha)$, the harmonic forward model defines a linear map from boundary pressure phasors to predicted edge-velocity phasors,

$$\hat{\mathbf{v}}^{(n)} = \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}^{(n)}_b. \tag{41}$$

Here $\widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)$ is the velocity-transfer matrix constructed from the harmonic network solve. The dependence on $(D_0, \alpha)$ enters through the edge distensibilities, compliances, hydraulic wavenumbers, and admittances. For fixed $(D_0, \alpha)$, however, the model remains linear in the boundary pressure phasors.

## 5.1 Least-Squares Formulation

Given measured velocity harmonics $\hat{\mathbf{v}}^{(n)}_{\text{obs}}$, we define the residual for harmonic $n$ as

$$\mathbf{r}^{(n)}_v = \hat{\mathbf{v}}^{(n)}_{\text{obs}} - \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}^{(n)}_b. \tag{42}$$

A natural inverse problem is to minimize the weighted squared residual,

$$\mathcal{J}\left(D_0, \alpha, \{\hat{\mathbf{P}}^{(n)}_b\}_{n\in\mathcal{N}}\right) = \sum_{n\in\mathcal{N}} \left\| \hat{\mathbf{v}}^{(n)}_{\text{obs}} - \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}^{(n)}_b \right\|^2_{\mathbf{W}^{(n)}_v}, \tag{43}$$

where $\mathcal{N}$ is the set of harmonics included in the fit, $\mathbf{W}^{(n)}_v$ is a positive semi-definite weighting matrix, and

$$\|\mathbf{x}\|^2_{\mathbf{W}} = \mathbf{x}^H\mathbf{W}\mathbf{x}.$$

The superscript $H$ denotes the conjugate transpose. If the measurement noise is independent across edges, $\mathbf{W}^{(n)}_v$ may be chosen as a diagonal matrix. For example, if edge $e$ has velocity-noise variance $\sigma^2_{v,e,n}$, then

$$\mathbf{W}^{(n)}_v = \operatorname{diag}\left(\frac{1}{\sigma^2_{v,1,n}}, \frac{1}{\sigma^2_{v,2,n}}, \dots, \frac{1}{\sigma^2_{v,N_e,n}}\right). \tag{44}$$

### 5.1.1 Ordinary Least Squares

In the ordinary least-squares case, the measurement variance is assumed to be the same for every measured edge within a harmonic,

$$\sigma^2_{v,1,n} = \sigma^2_{v,2,n} = \cdots = \sigma^2_{v,N_e,n} = \sigma^2_{v,n}.$$

Then $\mathbf{W}^{(n)}_v$ is proportional to the identity matrix,

$$\mathbf{W}_v^{(n)} = \frac{1}{\sigma_{v,n}^2}\mathbf{I},$$

and the scalar factor $1/\sigma_{v,n}^2$ does not change the least-squares minimizer for that harmonic. Therefore, when all edges are weighted equally, the weighted norm may be replaced by the ordinary Euclidean norm.

### 5.2 Profiling Out Boundary Pressures

The boundary pressure phasors are nuisance variables: they are required to drive the network, but they are not the primary quantities of interest. For fixed $(D_0, \alpha)$, the objective is quadratic in $\hat{\mathbf{P}}_b^{(n)}$, so the optimal boundary pressure phasors can be solved analytically by weighted complex least squares.

For each harmonic $n$,

$$\hat{\mathbf{P}}_b^{(n)}(D_0, \alpha) = \left[\widetilde{\mathbf{T}}^{(n)H}\mathbf{W}_v^{(n)}\widetilde{\mathbf{T}}^{(n)}\right]^{-1}\widetilde{\mathbf{T}}^{(n)H}\mathbf{W}_v^{(n)}\hat{\mathbf{v}}_{\text{obs}}^{(n)}, \tag{45}$$

where $\widetilde{\mathbf{T}}^{(n)} = \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)$. If the normal matrix is ill-conditioned, the inverse may be replaced by a pseudoinverse or a regularized least-squares solve.

Substituting equation 45 into equation 43 yields the profiled objective

$$\mathcal{J}_{\text{prof}}(D_0, \alpha) = \sum_{n\in\mathcal{N}} \left\| \hat{\mathbf{v}}_{\text{obs}}^{(n)} - \widetilde{\mathbf{T}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}_b^{(n)}(D_0, \alpha) \right\|_{\mathbf{W}_v^{(n)}}^2. \tag{46}$$

The distensibility parameters are then estimated by

$$(\hat{D}_0, \hat{\alpha}) = \arg\min_{D_0,\alpha} \mathcal{J}_{\text{prof}}(D_0, \alpha). \tag{47}$$

This formulation separates the nonlinear and linear parts of the inverse problem. The distensibility parameters $(D_0, \alpha)$ enter nonlinearly because they determine the hydraulic wavenumbers and admittances. The boundary pressure phasors enter linearly and can therefore be estimated exactly for each trial pair $(D_0, \alpha)$.

### 5.3 Special Cases for the Radius-Dependence Exponent $\alpha$

The exponent $\alpha$ controls how the distensibility varies with vessel radius,

$$D_e = D_0\left(\frac{R_e}{R_0}\right)^\alpha.$$

Since

$$D_{\text{hyd},e} = \frac{1}{r_e c_e} = \frac{R_0^\alpha}{8\mu D_0}R_e^{2-\alpha},$$

the hydraulic diffusivity scales as

$$D_{\text{hyd},e} \propto R_e^{2-\alpha}.$$

Equivalently, the harmonic hydraulic wavenumber satisfies

$$k_e^{(n)} = \left(\frac{in\omega_0}{D_{\mathrm{hyd},e}}\right)^{1/2} \propto R_e^{(\alpha-2)/2}.$$

Thus, $\alpha = 2$ is a natural threshold: it separates the regime where larger vessels have smaller hydraulic wavenumber from the regime where larger vessels have larger hydraulic wavenumber.

#### 5.3.1 Case 1: Constant Distensibility, $\alpha = 0$

Setting $\alpha = 0$ gives $D_e = D_0$, the constant-distensibility model. In this case, $D_{\mathrm{hyd},e} \propto R_e^2$, and $k_e^{(n)} \propto R_e^{-1}$. Therefore, larger vessels have larger hydraulic diffusivity and smaller hydraulic wavenumber. Physically, this means that larger vessels transmit pulsatile pressure variations farther, with less attenuation and less phase accumulation per unit length.

The profiled objective reduces to a one-dimensional function of $D_0$,

$$\mathcal{J}_{\mathrm{prof}}(D_0; \alpha = 0) = \sum_{n\in\mathcal{N}} \left\| \hat{\mathbf{v}}_{\mathrm{obs}}^{(n)} - \widetilde{\mathbf{T}}^{(n)}(D_0, 0)\hat{\mathbf{P}}_b^{(n)}(D_0, 0) \right\|_{\mathbf{W}_v^{(n)}}^2 .$$

The corresponding estimate is

$$\hat{D}_0 = \arg\min_{D_0} \mathcal{J}_{\mathrm{prof}}(D_0; \alpha = 0).$$

#### 5.3.2 Case 2: Radius-Independent Hydraulic Diffusivity, $\alpha = 2$

Setting $\alpha = 2$ gives $D_{\mathrm{hyd},e} \propto R_e^0$ and $k_e^{(n)} \propto R_e^0$. Thus, the hydraulic diffusivity and the harmonic wavenumber are independent of vessel radius. Larger and smaller vessels have the same attenuation and phase accumulation per unit length, although their steady conductances and velocity conductances still differ through their dependence on radius.

The profiled objective is

$$\mathcal{J}_{\mathrm{prof}}(D_0; \alpha = 2) = \sum_{n\in\mathcal{N}} \left\| \hat{\mathbf{v}}_{\mathrm{obs}}^{(n)} - \widetilde{\mathbf{T}}^{(n)}(D_0, 2)\hat{\mathbf{P}}_b^{(n)}(D_0, 2) \right\|_{\mathbf{W}_v^{(n)}}^2 ,$$

with estimate

$$\hat{D}_0 = \arg\min_{D_0} \mathcal{J}_{\mathrm{prof}}(D_0; \alpha = 2).$$

#### 5.3.3 Case 3: Larger Vessels Transmit Pulsatility Farther, $\alpha < 2$

When $\alpha < 2$, the $D_{hyd,e}$ scales positively with and $k_e^{(n)}$ scales negatively with $R_e$. Therefore, larger vessels have larger hydraulic diffusivity and smaller harmonic wavenumber. Physically, larger vessels attenuate pulsatile pressure variations less strongly and accumulate less phase lag per unit length. This is the regime in which larger-caliber vessels transmit pulsatility farther than smaller-caliber vessels.

#### 5.3.4 Case 4: Larger Vessels Attenuate Pulsatility More Strongly, $\alpha > 2$

When $\alpha > 2$ the $D_{hyd,e}$ scales negatively with and $k_e^{(n)}$ scales positively with $R_e$. Therefore, larger vessels have smaller hydraulic diffusivity and have larger harmonic wavenumber. Physically, larger vessels attenuate pulsatile pressure variations more strongly and accumulate more phase lag per unit length.

# 6 Joint Inference Methods

## 6.1 Deterministic Linear Solver

For fixed $(D_0, \alpha)$, the pulsatile network equations define a linear relationship between the unknown boundary pressure phasors and the predicted edge-velocity phasors. This observation motivates a deterministic inverse formulation in which the boundary pressures are treated as nuisance variables and eliminated by linear least squares.

### 6.1.1 Solving Each Tile Independently

We first consider a tile-wise inverse problem. Let $\tau$ denote a spatial tile, and let $\hat{\mathbf{v}}^{(n)}_{\text{obs},\tau}$ be the vector of measured edge-velocity phasors in tile $\tau$ at harmonic $n$. The harmonic forward model defines a linear map

$$\hat{\mathbf{v}}^{(n)}_{\tau} = \tilde{\mathbf{T}}^{(n)}_{\tau}(D_0, \alpha)\hat{\mathbf{P}}^{(n)}_{b,\tau}, \tag{48}$$

where $\hat{\mathbf{P}}^{(n)}_{b,\tau}$ contains the unknown pressure phasors at the boundary nodes of the tile. The transfer matrix $\tilde{\mathbf{T}}^{(n)}_{\tau}(D_0, \alpha)$ is obtained by assembling the harmonic admittance matrix for the tile, solving the interior pressure problem for each boundary pressure basis vector, and computing the resulting edge-flow phasors.

For fixed $(D_0, \alpha)$, the unknown boundary pressure phasors enter linearly. Therefore, for each harmonic, they can be estimated by weighted least squares,

$$\hat{\mathbf{P}}^{(n)}_{b,\tau}(D_0, \alpha) = \arg\min_{\mathbf{p}} \left| \mathbf{W}^{(n)1/2}_{v,\tau} \left[ \hat{\mathbf{v}}^{(n)}_{\text{obs},\tau} - \tilde{\mathbf{T}}^{(n)}_{\tau}(D_0, \alpha)\mathbf{p} \right] \right|_2^2, \tag{49}$$

where $\mathbf{W}^{(n)}_{v,\tau}$ is a diagonal matrix of measurement weights. The closed-form solution is

$$\hat{\mathbf{P}}^{(n)}_{b,\tau}(D_0, \alpha) = \left[ \mathbf{T}^{(n)H}_{\tau} \mathbf{W}^{(n)}_{v,\tau} \mathbf{T}^{(n)}_{\tau} \right]^{-1} \mathbf{T}^{(n)H}_{\tau} \mathbf{W}^{(n)}_{v,\tau} \hat{\mathbf{v}}^{(n)}_{\text{obs},\tau}, \tag{50}$$

where $H$ denotes the conjugate transpose. Substituting this solution back into the residual gives the profiled objective

$$\chi^2_{\tau,\text{prof}}(D_0, \alpha) = \sum_{n \geq 1} \left| \mathbf{W}^{(n)1/2}_{v,\tau} \left[ \hat{\mathbf{v}}^{(n)}_{\text{obs},\tau} - \tilde{\mathbf{T}}^{(n)}_{\tau}(D_0, \alpha)\hat{\mathbf{P}}^{(n)}_{b,\tau}(D_0, \alpha) \right] \right|_2^2. \tag{51}$$

The zero-frequency component may be included to constrain the mean pressure field or boundary forcing, but the distensibility information is carried primarily by the pulsatile harmonics because the steady Poiseuille network does not depend on vessel compliance. The tile-wise estimate is then obtained by minimizing the profiled objective,

$$(\hat{D}_0, \hat{\alpha})_{\tau} = \arg\min_{D_0, \alpha} \chi^2_{\tau,\text{prof}}(D_0, \alpha).$$

The minimization can be performed by scanning a grid over $D_0$ and, when applicable, $\alpha$. The shape of $\chi^2_{\tau,\text{prof}}(D_0, \alpha)$ provides a diagnostic for practical identifiability. A sharp minimum indicates that the measured harmonics constrain the wall-law parameters, whereas a broad or nearly flat profile indicates that multiple values of $(D_0, \alpha)$ explain the measurements equally well.

### 6.1.2 Solving the Whole Mosaic

The same deterministic formulation can be applied to the full vascular mosaic. Instead of treating each tile as an independent subgraph with its own artificial boundary nodes, the entire vascular network is assembled into a single harmonic admittance problem. Let $\hat{\mathbf{v}}_{\text{obs}}^{(n)}$ denote the measured edge-velocity phasors across the mosaic. For fixed parameters $(D_0, \alpha)$, the global forward model is

$$\hat{\mathbf{v}}^{(n)} = \tilde{\mathbf{T}}_{\text{mos}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}_b^{(n)}, \tag{52}$$

where $\hat{\mathbf{P}}_b^{(n)}$ now represents pressure phasors at the external boundary nodes of the mosaic rather than at the boundary of each tile. The corresponding profiled least-squares objective is

$$\chi^2_{\text{mos,prof}}(D_0, \alpha) = \sum_{n\geq 1} \left| \mathbf{W}_{v,\text{mos}}^{(n)1/2} \left[ \hat{\mathbf{Q}}_{\text{obs}}^{(n)} - \tilde{\mathbf{T}}_{\text{mos}}^{(n)}(D_0, \alpha)\hat{\mathbf{P}}_b^{(n)}(D_0, \alpha) \right] \right|_2^2, \tag{53}$$

with

$$\hat{\mathbf{P}}_b^{(n)}(D_0, \alpha) = \left[ \tilde{\mathbf{T}}_{\text{mos}}^{(n)H} \mathbf{W}_{v,\text{mos}}^{(n)} \tilde{\mathbf{T}}_{\text{mos}}^{(n)} \right]^{-1} \tilde{\mathbf{T}}_{\text{mos}}^{(n)H} \mathbf{W}_{v,\text{mos}}^{(n)} \hat{\mathbf{v}}_{\text{obs}}^{(n)}. \tag{54}$$

The global estimate is

$$(\hat{D_0}, \hat{\alpha})_{\text{mos}} = \arg\min_{D_0,\alpha} \chi^2_{\text{mos,prof}}(D_0, \alpha).$$

The physical distinction between the whole-mosaic and tile-wise formulations is that the whole-mosaic solve couples all measured vessels through a shared pressure field on the full graph. Information is not pooled by simply concatenating independent residuals; rather, coupling occurs through the global admittance matrix, which encodes the vascular topology, vessel conductances, and compliant-wall admittances. This formulation is therefore most appropriate when the entire mosaic can be represented as a connected network with well-defined external boundary conditions.

The whole-mosaic formulation can improve practical identifiability by increasing the number of measured edges and reducing the number of artificial tile-boundary forcing variables. However, it also imposes a stronger modeling assumption: the selected wall law must be valid over the full mosaic, or any spatial variation in wall properties must be explicitly parameterized.

## 6.2 Bayesian Solver

The deterministic solver treats the boundary pressure phasors as unknown quantities to be optimized. The Bayesian formulation instead treats these boundary phasors as nuisance variables and integrates over them. This distinction is important because uncertainty in the unknown boundary forcing is propagated into the posterior distribution over the distensibility parameters.

### 6.2.1 Solving Each Tile Independently

For a tile $\tau$, the measured data $\mathbf{y}_\tau$ and boundary forcing vector $\mathbf{b}_\tau$ are stacked into a real-valued vector

$$\mathbf{y}_\tau = \begin{bmatrix} \bar{\mathbf{v}}_\tau \\ \Re\tilde{\mathbf{v}}_\tau^{(1)} \\ \Im\tilde{\mathbf{v}}_\tau^{(1)} \end{bmatrix}, \qquad \mathbf{b}_\tau = \begin{bmatrix} \mathbf{b}_\tau^{(0)} \\ \Re\mathbf{b}_\tau^{(1)} \\ \Im\mathbf{b}_\tau^{(1)} \end{bmatrix},$$

where $\bar{\mathbf{v}}_\tau$ is the steady component velocity and $\hat{\mathbf{v}}_\tau^{(1)}$ is the first-harmonic velocity phasor. Additional harmonics may be added, and the formulation below remains the same.

For fixed $(D_0, \alpha)$, the real-valued forward model is

$$\mathbf{y}_\tau = \mathbf{T}_\tau(D_0, \alpha)\mathbf{b}_\tau + \boldsymbol{\varepsilon}_\tau. \tag{55}$$

The measurement noise is modeled as Gaussian,

$$\boldsymbol{\varepsilon}_\tau \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{n,\tau}),$$

where $\boldsymbol{\Sigma}_{n,\tau}$ may include separate noise scales for the DC and harmonic channels, as well as relative edge weights based on measurement quality or signal-to-noise ratio.

The distensibility parameter is assigned a prior distribution. For a positive scale parameter, a log-normal prior is natural,

$$\log D_0 \sim \mathcal{N}(\mu_D, \tau_D^2).$$

If the radius exponent is inferred, it may be assigned a Gaussian prior,

$$\alpha \sim \mathcal{N}(\mu_\alpha, s_\alpha^2).$$

The boundary forcing is also assigned a Gaussian prior,

$$\mathbf{b}_\tau \sim \mathcal{N}(\mathbf{b}_0, \mathbf{P}_0).$$

Conditional on $(D_0, \alpha)$, the forward model is linear in $\mathbf{b}_\tau$, and both the likelihood and boundary-forcing prior are Gaussian. Therefore, the boundary forcing can be marginalized analytically:

$$p(\mathbf{y}_\tau \mid D_0, \alpha) = \int p(\mathbf{y}_\tau \mid D_0, \alpha, \mathbf{b}_\tau) p(\mathbf{b}_\tau) d\mathbf{b}_\tau. \tag{56}$$

The resulting marginal distribution is Gaussian,

$$\mathbf{y}_\tau \mid D_0, \alpha \sim \mathcal{N}\left(\tilde{\mathbf{T}}_\tau(D_0, \alpha)\mathbf{b}_0, \mathbf{C}_\tau(D_0, \alpha)\right), \tag{57}$$

with covariance

$$\mathbf{C}_\tau(D_0, \alpha) = \boldsymbol{\Sigma}_{n,\tau} + \tilde{\mathbf{T}}_\tau(D_0, \alpha)\mathbf{P}_0\tilde{\mathbf{T}}_\tau(D_0, \alpha)^T.$$

The first term represents measurement uncertainty, while the second term represents uncertainty in the unknown boundary forcing propagated through the forward model. The marginal log likelihood is therefore

$$\begin{aligned}\log p(\mathbf{y}_\tau \mid D_0, \alpha) = &-\frac{1}{2}\log|\mathbf{C}_\tau(D_0, \alpha)| - \frac{1}{2}\mathbf{r}_\tau(D_0, \alpha)^\top \mathbf{C}_\tau(D_0, \alpha)^{-1}\mathbf{r}_\tau(D_0, \alpha) \\ &+ \text{const}\end{aligned} \tag{58}$$

where

$$\mathbf{r}_\tau(D_0, \alpha) = \mathbf{y}_\tau - \tilde{\mathbf{T}}_\tau(D_0, \alpha)\mathbf{b}_0.$$

The quantity evaluated in the Bayesian solver is the corresponding log posterior surface,

$$
\begin{aligned}
\ell_\tau(D_0, \alpha) &= \log p(\mathbf{y}_\tau \mid D_0, \alpha) + \log p(D_0) + \log p(\alpha) \\
&= -\frac{1}{2} \log |\mathbf{C}_\tau(D_0, \alpha)| - \frac{1}{2} \mathbf{r}_\tau(D_0, \alpha)^\top \mathbf{C}_\tau(D_0, \alpha)^{-1} \mathbf{r}_\tau(D_0, \alpha) \\
&\quad + \log p(D_0) + \log p(\alpha) + \text{const.}
\end{aligned} \tag{59}
$$

This surface is evaluated on a grid over $\log D_0$ and, when applicable, $\alpha$. If $\alpha$ is treated as a nuisance parameter, the marginal posterior for $D_0$ is obtained by summing over the $\alpha$ grid,

$$
p(D_0^{(k)} \mid \mathbf{y}_\tau) \propto \sum_m \exp\left[\ell_\tau(D_0^{(k)}, \alpha^{(m)})\right] \Delta\alpha, \tag{60}
$$

followed by normalization over the $D_0$ grid. If $\alpha$ is fixed, this reduces to a one-dimensional posterior curve over $D_0$. The posterior mode is used as a point estimate, and credible intervals quantify uncertainty in the inferred distensibility parameters. A tile is considered practically identifiable when the posterior departs from the prior and exhibits a well-defined interior mode. Conversely, a broad posterior or a posterior that closely resembles the prior indicates that the measured data do not strongly constrain the distensibility in that tile.

### 6.2.2 Solving the Whole Mosaic

The Bayesian formulation can also be applied to the full vascular mosaic. The measured data $\mathbf{y}_{\text{mos}}$ and boundary forcing vector $\mathbf{b}_{\text{mos}}$ are stacked into a real-valued vector

$$
\mathbf{y}_{\text{mos}} = \begin{bmatrix} \bar{\mathbf{v}}_{\text{mos}} \\ \Re\tilde{\mathbf{v}}_{\text{mos}}^{(1)} \\ \Im\tilde{\mathbf{v}}_{\text{mos}}^{(1)} \end{bmatrix}, \qquad \mathbf{b}_{\text{mos}} = \begin{bmatrix} \mathbf{b}_{\text{mos}}^{(0)} \\ \Re\mathbf{b}_{\text{mos}}^{(1)} \\ \Im\mathbf{b}_{\text{mos}}^{(1)} \end{bmatrix}.
$$

Additional harmonics may be added, and the formulation below remains the same. The forward operator $\tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)$ is constructed from the harmonic admittance matrix of the entire mosaic. The model is

$$
\mathbf{y}_{\text{mos}} = \tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)\mathbf{b}_{\text{mos}} + \boldsymbol{\varepsilon}_{\text{mos}}, \tag{61}
$$

where $\mathbf{b}_{\text{mos}}$ contains the unknown forcing amplitudes at the external boundary nodes of the full mosaic. As in the tile-wise case, the boundary forcing is assigned a Gaussian prior,

$$
\mathbf{b}_{\text{mos}} \sim \mathcal{N}(\mathbf{b}_0, \mathbf{P}_0),
$$

and the measurement noise is modeled as

$$
\boldsymbol{\varepsilon}_{\text{mos}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{n,\text{mos}}).
$$

Marginalizing over the unknown boundary forcing gives

$$
\mathbf{y}_{\text{mos}} \mid D_0, \alpha \sim \mathcal{N}\left(\tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)\mathbf{b}_0, \mathbf{C}_{\text{mos}}(D_0, \alpha)\right), \tag{62}
$$

where

$$
\mathbf{C}_{\text{mos}}(D_0, \alpha) = \boldsymbol{\Sigma}_{n,\text{mos}} + \tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)\mathbf{P}_0\tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)^T.
$$

The Bayesian whole-mosaic solver evaluates the log posterior surface

$$
\begin{aligned}
\ell_{\text{mos}}(D_0, \alpha) &= \log p(\mathbf{y}_{\text{mos}} \mid D_0, \alpha) + \log p(D_0) + \log p(\alpha) \\
&= -\frac{1}{2} \log |\mathbf{C}_{\text{mos}}(D_0, \alpha)| - \frac{1}{2} \mathbf{r}_{\text{mos}}(D_0, \alpha)^\top \mathbf{C}_{\text{mos}}(D_0, \alpha)^{-1} \mathbf{r}_{\text{mos}}(D_0, \alpha) \\
&\quad + \log p(D_0) + \log p(\alpha) + \text{const.}
\end{aligned} \tag{63}
$$

where

$$
\mathbf{r}_{\text{mos}}(D_0, \alpha) = \mathbf{y}_{\text{mos}} - \tilde{\mathbf{T}}_{\text{mos}}(D_0, \alpha)\mathbf{b}_0.
$$

This surface is evaluated on a grid over $\log D_0$ and, when applicable, $\alpha$. If $\alpha$ is treated as a nuisance parameter, the marginal posterior for $D_0$ is obtained by summing over the $\alpha$ grid,

$$
p(D_0^{(k)} \mid \mathbf{y}_{\text{mos}}) \propto \sum_m \exp\left[\ell_{\text{mos}}(D_0^{(k)}, \alpha^{(m)})\right] \Delta\alpha, \tag{64}
$$

followed by normalization over the $D_0$ grid. If $\alpha$ is fixed, this reduces to a one-dimensional posterior curve of $D_0$. The posterior mode provides the whole-mosaic point estimate, while the width of the posterior surface quantifies uncertainty in the inferred distensibility parameters.

This formulation differs from the tile-wise Bayesian solver in two ways. First, the pressure field is inferred on the full mosaic, so spatial coupling is imposed through the global harmonic admittance matrix rather than through independent tile boundary conditions. Second, the number of nuisance boundary forcing variables can be reduced because only the external boundaries of the mosaic require forcing variables, rather than every artificial tile cut. The whole-mosaic Bayesian solver is therefore a more strongly coupled inverse problem. It can improve practical identifiability by allowing all measured edges to constrain a shared distensibility model. However, this benefit depends on the validity of the assumed spatial structure of the distensibility model. If a single pair $(D_0, \alpha)$ is used for the entire mosaic, then the model assumes that radius dependence captures the dominant variation in distensibility. If substantial regional variation is expected, the parameter vector can be expanded to include tile-specific or region-specific parameters, with additional regularization or hierarchical priors used to prevent overfitting.

When the full mosaic is not solved as a single connected graph, an alternative is to combine the tile-wise marginal likelihoods. In that case, the pooled log posterior is

$$
\ell_{\text{pool}}(D_0, \alpha) = \log p(D_0) + \log p(\alpha) + \sum_\tau \log p(\mathbf{y}_\tau \mid D_0, \alpha). \tag{65}
$$

This pooled formulation uses all tiles to constrain a shared distensibility model, but it does not enforce pressure continuity between neighboring tiles. It should therefore be interpreted as a composite likelihood over tile-wise inverse problems rather than a fully coupled whole-mosaic solve.

# 7 Staged Graph Neural Network Inference

## 7.1 Motivation for Graph Neural Networks

The deterministic and Bayesian solvers described above use the harmonic admittance matrix to couple pressures and flows throughout the network. In the whole-mosaic formulation, this coupling is global: a perturbation at one node can, in principle, influence the inferred pressure field throughout the connected graph. Physically, however, the influence of distant nodes is strongly attenuated by viscous dissipation and vessel compliance. In the low-inertia regime considered here, pulsatile pressure disturbances spread diffusively rather than as undamped waves, so the contribution of a

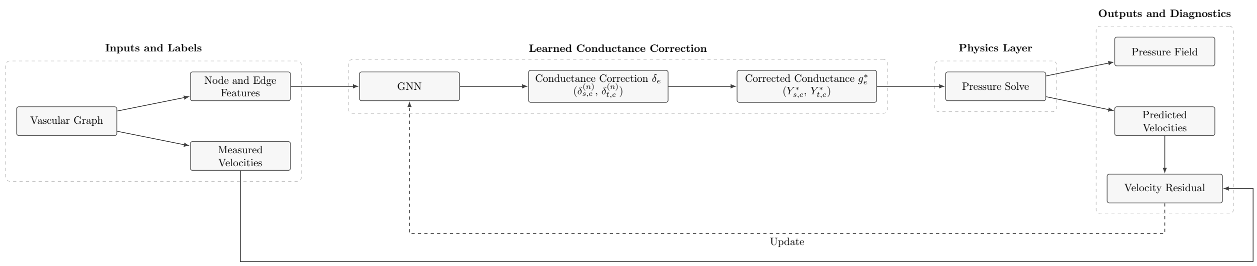


Figure 2: Physics-informed GNN workflow. The vascular graph provides measured velocities as well as node and edge features. The GNN estimates conductance corrections, which are subsequently used to correct conductances. The physics layer performs a pressure solve to generate a pressure field and predicted velocities. The predicted velocities are compared with the measured velocities to compute the velocity residual, which guides how the weights of the GNN are updated.

distant vessel decreases with hydraulic distance. Thus, although the full inverse problem is globally coupled, its dominant interactions are expected to be local.

This observation motivates a staged graph neural network (GNN) formulation (see Figure 2). Rather than allowing every node to interact with every other node through a dense global inverse map, the GNN restricts information exchange to a finite graph neighborhood. After $K$ message-passing layers, the representation of an edge depends only on nodes and vessels within $K$ graph hops. The choice of $K$ therefore provides an explicit and interpretable control on the spatial extent of interaction. This locality is a strength of the GNN formulation: it reflects the physical expectation that nearby vessels provide the most relevant information, while distant vessels contribute progressively weaker corrections.

## 7.2 The Correction Factor

The role of the GNN is to perform *edge-wise conductance correction*. The Poiseuille hydraulic conductance scaled by area is given in equation 30. The conductance corrections account for non-ideal flow, which may arise from bifurcations, segmentation uncertainty, or unresolved network structure. Thus, we introduce a correction $\delta_e$ estimated by the GNN,

$$g_e^* = \tilde{g}_e \exp(\delta_e) = \frac{R_e^2}{8\mu L_e} \exp(\delta_e) \tag{66}$$

The velocity along a vessel is calculated as

$$v_e^{(0)} = g_e^* (P_i^{(0)} - P_j^{(0)}) \tag{67}$$

Similar formulations can be made for the corrections to the velocity admittances,

$$Y_{s,e}^{*(n)} = \tilde{Y}_{s,e}^{(n)} \exp(\delta_{s,e}^{(n)}) \qquad Y_{t,e}^{*(n)} = \tilde{Y}_{t,e}^{(n)} \exp(\delta_{t,e}^{(n)}) \tag{68}$$

The velocity harmonics for $n \geq 1$ are calculated as

$$\hat{v}_e^{(n)} = Y_{s,e}^{*(n)}(D_0, \alpha)\hat{P}_i^{(n)} - Y_{t,e}^{*(n)}(D_0, \alpha)\hat{P}_j^{(n)} \tag{69}$$

### 7.2.1 Correction Factor Interpretability

The learned correction factor $\delta_e$ (as well as $\delta_{s,e}^{(n)}$ and $\delta_{t,e}^{(n)}$ if applicable) can be regressed against measured physical and geometric quantities, such as vessel radius, vessel length, endpoint degree, local branching structure, or measurement quality. This provides a post-hoc explanatory analysis of the model output by identifying which features are associated with deviations from the ideal Poiseuille conductance. Because this analysis is performed after training, it should be interpreted as an association between the learned correction field and measured vessel properties, rather than as direct evidence that the GNN uses each feature causally.

A related but distinct analysis is linear probing. In linear probing, the trained GNN is frozen and a simple linear model is trained on an intermediate hidden representation, such as an edge embedding after a given message-passing layer. The goal is to determine whether a physical quantity is encoded in the latent representation in a linearly accessible form. Thus, regressing the correction factor against input features explains the final correction output, whereas linear probing examines what information is represented internally by the GNN before the final prediction head.

### 7.2.2 Correction Factor Relationship to Distensibility Inference

Using only equations 66 and 67, the GNN learns corrections to the steady Poiseuille conductance, rather than corrections to the pulsatile admittance model. This produces a corrected conductance field and a corresponding DC pressure estimate. Since the GNN is not trained on the harmonic components, it does not directly observe the attenuation or phase shifts that are used to identify the distensibility parameters $D_0$ and $\alpha$. The harmonic inverse problem therefore remains responsible for estimating distensibility.

The motivation for the DC-only GNN is to reduce uncertainty in the steady pressure field before fitting the pulsatile model. By providing a corrected pressure estimate from the steady flow data, the GNN supplies prior information for the harmonic inverse problem. This can reduce the number or flexibility of nuisance pressure variables that must be inferred jointly with $D_0$ and $\alpha$, while preserving a clear separation between steady pressure estimation and harmonic distensibility inference.

As an ablation, the GNN can also be trained using harmonic information by also using equations 68 and 69. These harmonic-informed models test whether allowing the GNN to see pulsatile flow components improves reconstruction accuracy or stabilizes downstream inference. However, they also introduce a potential identifiability concern. Because harmonic attenuation and phase shifts are the same signals used to infer vessel compliance, a harmonic-informed GNN may absorb effects that would otherwise be attributed to $D_0$ and $\alpha$. For this reason, harmonic-informed GNNs should be interpreted as ablation studies that evaluate the tradeoff between predictive performance and interpretability, rather than as the cleanest formulation for distensibility inference.

## 7.3 Features

### 7.3.1 Node Features

For each node $i$, the features are

$$\mathbf{x}_i = [a_i^{\text{norm}}, b_i^{\text{norm}}, d_i, s_i, \mathbf{1}_{\text{source}}(i), \mathbf{1}_{\text{sink}}(i)]$$

where $(a_i^{\text{norm}}, b_i^{\text{norm}})$ are the dimensionless spatial coordinates of the node, $d_i$ is the node degree, $s_i$ is the prescribed source or sink velocity, and the indicator variables identify whether the node is a source, sink, or neither (i.e., interior). The node degree identifies bifurcations and high-connectivity regions, while the source/sink features specify externally driven boundary nodes.

The dimensionless spatial coordinates are calculated by

$$\left[a_i^{\text{norm}},\ b_i^{\text{norm}}\right] = \frac{\mathbf{R}_{\text{emb}}\left(\left[a_i,\ b_i\right] - \left[a_{\text{ref}},\ b_{\text{ref}}\right]\right)}{L_{\text{ref}}}.$$

Here, $(a_i, b_i)$ are the original whole-mosaic pixel coordinates, $(a_{\text{ref}}, b_{\text{ref}})$ is a reference location, $L_{\text{ref}}$ is a characteristic length scale, and $\mathbf{R}_{\text{emb}}$ is an optional embryo-specific rotation matrix used to align embryos into a common coordinate frame. In the simplest implementation, $(a_{\text{ref}}, b_{\text{ref}})$ may be chosen as the source or heart location, or the centroid of all source nodes when multiple sources are present,

$$\left[a_{\text{ref}},\ b_{\text{ref}}\right] = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}}\left[a_i,\ b_i\right],$$

where $\mathcal{S}$ is the set of source nodes. The reference length may be chosen as the mosaic diagonal,

$$L_{\text{ref}} = \sqrt{W_{\text{mos}}^2 + H_{\text{mos}}^2},$$

where $W_{\text{mos}}$ and $H_{\text{mos}}$ are the mosaic width and height in pixels. If no rotational alignment is applied, then $\mathbf{R}_{\text{emb}}$ is set to the identity matrix.

Optionally, harmonic features may be added:

$$\mathbf{x}_{i,\text{harmonic}} = [\Re\hat{H}_i^{(1)}, \Im\hat{H}_i^{(1)}, \Re\hat{H}_i^{(2)}, \Im\hat{H}_i^{(2)}]$$

where $\hat{H}_i^{(n)}$ is the complex boundary harmonic forcing phasor for harmonic $n$. These harmonic node features are not included in the primary DC-only model, but may be used in ablation studies.

### 7.3.2 Edge Features

For edge $e$ connecting nodes $i$ and $j$, the features are

$$\mathbf{x}_e = \mathbf{x}_{ij} = [\log R_e, \log L_e, \log\tilde{g}_e]$$

$\log\tilde{g}_e$ provides the GNN with the steady Poiseuille velocity scaling, so the network can focus on learning multiplicative corrections to the idealized conductance rather than rediscovering the baseline radius–length dependence from data.

## 7.4 Architecture

The GNN follows an encode–message passing–decode structure. First, node and edge features are embedded into a latent space of dimension $H$ using independent multilayer perceptrons,

$$\mathbf{h}_i^{(0)} = \epsilon_V(\mathbf{x}_i), \qquad \mathbf{h}_{ij}^{(0)} = \epsilon_E(\mathbf{x}_{ij}), \tag{70}$$

where $\epsilon_V$ and $\epsilon_E$ are the node and edge encoders, respectively. Although each physical vessel is undirected for the purpose of conductance, message passing is implemented on directed edges so that information can be propagated separately in both directions. For message-passing layer $k = 0, \ldots, K-1$, the directed edge embedding is updated using the current embeddings of the sender node, receiver node, and edge:

$$\mathbf{m}_{ij}^{(k)} = \phi_E^{(k)}\left([\mathbf{h}_i^{(k)}, \mathbf{h}_j^{(k)}, \mathbf{h}_{ij}^{(k)}]\right), \qquad \mathbf{h}_{ij}^{(k+1)} = \mathbf{h}_{ij}^{(k)} + \mathbf{m}_{ij}^{(k)}. \tag{71}$$

The residual connection stabilizes training and allows each layer to learn an incremental update to the current representation. Node embeddings are updated by aggregating incoming edge messages from neighboring nodes,

$$\bar{\mathbf{m}}_i^{(k)} = \sum_{j\in\mathcal{N}(i)} \mathbf{h}_{ji}^{(k+1)}, \tag{72}$$

followed by a node update

$$\mathbf{u}_i^{(k)} = \phi_V^{(k)}\left([\mathbf{h}_i^{(k)}, \bar{\mathbf{m}}_i^{(k)}]\right), \qquad \mathbf{h}_i^{(k+1)} = \mathbf{h}_i^{(k)} + \mathbf{u}_i^{(k)}. \tag{73}$$

Here, $\mathcal{N}(i)$ denotes the set of nodes adjacent to node $i$, and $\phi_E^{(k)}$ and $\phi_V^{(k)}$ are edge and node update networks. The summation aggregation makes the update invariant to the ordering of neighboring vessels. After $K$ message-passing layers, the final node and edge embeddings are decoded into an edge-wise conductance correction. For a edge $e = (i, j)$, the correction is symmetrized across the two directed orientations:

$$\begin{gathered}\tilde{\delta}_{ij} = \psi_\delta\left([\mathbf{h}_i^{(K)}, \mathbf{h}_j^{(K)}, \mathbf{h}_{ij}^{(K)}]\right), \qquad \tilde{\delta}_{ji} = \psi_\delta\left([\mathbf{h}_j^{(K)}, \mathbf{h}_i^{(K)}, \mathbf{h}_{ji}^{(K)}]\right), \\ \delta_e = \frac{1}{2}\left(\tilde{\delta}_{ij} + \tilde{\delta}_{ji}\right).\end{gathered} \tag{74}$$

This produces a single scalar correction for each vessel segment that does not depend on the arbitrary orientation used during message passing.

## 7.5 Conductance Correction and Physics Layer

The decoder predicts the conductance corrections $\delta_e$, which are then used to calculate corrected conductances $g_e^*$. The corrected conductances are assembled into the weighted graph Laplacian $\mathbf{L}^*$. For an edge $e = (i, j)$, the local contributions are

$$\begin{aligned} L_{ii}^* &\leftarrow L_{ii}^* + g_e^*, \qquad & L_{jj}^* &\leftarrow L_{jj}^* + g_e^*, \\ L_{ij}^* &\leftarrow L_{ij}^* - g_e^*, \qquad & L_{ji}^* &\leftarrow L_{ji}^* - g_e^*. \end{aligned}$$

The nodal pressure field is then obtained from the velocity-scaled network equation

$$\mathbf{L}^*\mathbf{P}^{(0)} = \mathbf{H}_v^{(0)}, \tag{75}$$

together with a pressure gauge. Here, $\mathbf{H}_v^{(0)}$ is the net mean velocities at each node. Finally, the predicted mean velocity is calculated using equation 67. The GNN is therefore trained end-to-end through the differentiable map

$$\{\mathbf{x}_i, \mathbf{x}_e\} \longrightarrow \delta_e \longrightarrow \mathbf{L}^* \longrightarrow \mathbf{P}^{(0)} \longrightarrow v_{\mathrm{pred},e}^{(0)} \tag{76}$$

### 7.6 Training Objective

Let $\mathcal{E}_{\text{obs}}$ denote the set of edges with measured mean velocities. The velocity loss is the weighted velocity residual

$$\mathcal{L}_{\text{vel}} = \frac{1}{\sum_{e\in\mathcal{E}_{\text{obs}}} w_e} \sum_{e\in\mathcal{E}_{\text{obs}}} w_e \left(v^{(0)}_{\text{obs},e} - v^{(0)}_{\text{pred},e}\right)^2, \tag{77}$$

where $w_e$ is an optional measurement-quality weight, such as an inverse variance or signal-to-noise weight. If all observed edges are weighted equally, $w_e = 1$.

Because the correction factor is intended to capture deviations from the idealized Poiseuille model rather than replace it entirely, we regularize the magnitude of the learned correction:

$$\mathcal{L}_\delta = \frac{1}{|\mathcal{E}|} \sum_{e\in\mathcal{E}} \delta_e^2. \tag{78}$$

Finally, to avoid unrealistically large pressure variations, we introduce a pressure loss:

$$\mathcal{L}_{\text{pressure}} = \frac{1}{|\mathcal{E}|} \sum_{e\in\mathcal{E}} (P_i^{(0)} - P_j^{(0)})^2 \tag{79}$$

The full training objective is

$$\mathcal{L}_{\text{GNN}} = \mathcal{L}_{\text{vel}} + \lambda_\delta \mathcal{L}_\delta + \lambda_{\text{pressure}} \mathcal{L}_{\text{pressure}} \tag{80}$$

### 7.7 Using the GNN Pressure Prior for Distensibility Inference

The fully trained GNN produces a predicted pressure field, $\mathbf{P}^{(0)}_{GNN}$. This pressure field is then applied to equations 53 or 64 to solve for the distensibility parameters $D_0$ and $\alpha$. In the latter case, a covariance matrix, $\mathbf{\Sigma}_P$, may be prescribed to adjust the informativeness of the prior.

The harmonic boundary pressure phasors remain nuisance variables in the distensibility inference, and the distensibility parameters are still inferred from the harmonic velocity residuals. This staging preserves interpretability. The GNN accounts for local steady-flow deviations from ideal Poiseuille behavior, while the harmonic inverse problem remains responsible for estimating distensibility.

### 7.8 Harmonic-Informed Ablation

As an ablation study, the same workflow (Figure 2) can be extended to predict corrections to the harmonic velocity admittances. For each harmonic $n \in \mathcal{N}_h$, the corrected harmonic velocity admittances from equation 68 are assembled into a complex harmonic nodal matrix $\mathbf{Y}^{*(n)}$. For an edge $e = (i, j)$, the local contributions are

$$\begin{aligned} Y^{*(n)}_{ii} &\leftarrow Y^{*(n)}_{ii} + Y^{*(n)}_{s,e}, \qquad & Y^{*(n)}_{jj} &\leftarrow Y^{*(n)}_{jj} + Y^{*(n)}_{s,e}, \\ Y^{*(n)}_{ij} &\leftarrow Y^{*(n)}_{ij} - Y^{*(n)}_{t,e}, \qquad & Y^{*(n)}_{ji} &\leftarrow Y^{*(n)}_{ji} - Y^{*(n)}_{t,e}. \end{aligned}$$

The harmonic pressure phasors are then obtained from

$$\mathbf{Y}^{*(n)} \hat{\mathbf{P}}^{(n)} = \hat{\mathbf{H}}^{(n)}_v, \tag{81}$$

together with a pressure gauge or boundary pressure constraint. Here, $\hat{\mathbf{H}}_v^{(n)}$ is the net harmonic velocities at each node. Once $\hat{\mathbf{P}}^{(n)}$ is obtained, the predicted edge-velocity phasors are computed using equation 69.

Let $\mathcal{E}_{\text{obs}}^{(n)}$ denote the set of edges with measured velocity phasors for harmonic $n$. The harmonic velocity loss is

$$\mathcal{L}_{\text{harm}} = \frac{1}{\sum_{n\in\mathcal{N}_h}\sum_{e\in\mathcal{E}_{\text{obs}}^{(n)}} w_{e,n}} \sum_{n\in\mathcal{N}_h} \sum_{e\in\mathcal{E}_{\text{obs}}^{(n)}} w_{e,n} \left| \hat{v}_{\text{obs},e}^{(n)} - \hat{v}_{\text{pred},e}^{(n)} \right|^2, \tag{82}$$

where $w_{e,n}$ is an optional harmonic-specific measurement weight. The complex residual may equivalently be written in terms of its real and imaginary components,

$$\left| \hat{v}_{\text{obs},e}^{(n)} - \hat{v}_{\text{pred},e}^{(n)} \right|^2 = \left( \Re\hat{v}_{\text{obs},e}^{(n)} - \Re\hat{v}_{\text{pred},e}^{(n)} \right)^2 + \left( \Im\hat{v}_{\text{obs},e}^{(n)} - \Im\hat{v}_{\text{pred},e}^{(n)} \right)^2 \tag{83}$$

The harmonic correction penalty is

$$\mathcal{L}_{\delta,\text{harm}} = \frac{1}{|\mathcal{N}_h||\mathcal{E}|} \sum_{n\in\mathcal{N}_h} \sum_{e\in\mathcal{E}} \left[ \left( \delta_{s,e}^{(n)} \right)^2 + \left( \delta_{t,e}^{(n)} \right)^2 \right]. \tag{84}$$

To avoid unrealistically large harmonic pressure variations, an analogous harmonic pressure loss may be added,

$$\mathcal{L}_{\text{pressure,harm}} = \frac{1}{|\mathcal{N}_h||\mathcal{E}|} \sum_{n\in\mathcal{N}_h} \sum_{e=(i,j)\in\mathcal{E}} \left| \hat{P}_i^{(n)} - \hat{P}_j^{(n)} \right|^2. \tag{85}$$

The harmonic-informed ablation objective is therefore

$$\mathcal{L}_{\text{GNN,harm}} = \mathcal{L}_{\text{GNN}} + \lambda_{\text{harm}}\mathcal{L}_{\text{harm}} + \lambda_{\delta,\text{harm}}\mathcal{L}_{\delta,\text{harm}} + \lambda_{\text{pressure,harm}}\mathcal{L}_{\text{pressure,harm}}. \tag{86}$$